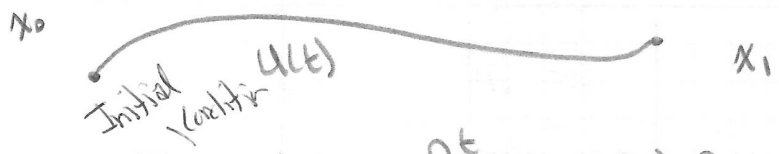
I Overview

Can we move between two points?



Is there a $u(t)$ that will get us from ~~that~~ x_0 to x_1.

$$x(t) = \underbrace{\Phi(t,t_0)}_{\text{State transition matrix}} x_0 + \underbrace{\int_{t_0}^{t} \Phi(t,\tau) B(\tau) u(\tau)\, d\tau}_{\text{Control Inputs}}$$

A) How to show two subspaces are equal

Sub-spaces are really just sets

V_1 is a subspace, V_2 is a subspace

$\Rightarrow V_1 = V_2$ iff $a \in V_1 \Leftrightarrow a \in V_2$

B) Reachable Subspace

All the possible states that we can get to if we start at the zero state at t_0 and move until t_1
$R[t_0, t_1]$

C) Controllable Subspaces

All the possible states that we can start at can be driven to 0. $C[t_0, t_1]$

These are easy to define, but hard to come up with.

We will show that these subspaces are actually the image space of two matrices:

Reachability Grammian $W_R(t_0, t_1)$

Controllability Grammian $W_C(t_0, t_1)$

II Reachable and Controllable Subspaces

A) Reachable Subspace

Given two times $t_0 \leq t_1$ the reachable subspace on interval $[t_0, t_1]$

$$R[t_0, t_1] = \{ x \in \mathbb{R}^n \mid \exists\, u(\cdot) \in \mathcal{U}(t_0, t_1);\ x_1 = \int_{t_0}^{t} \Phi(t, \tau) B(\tau) u(\tau)\, d\tau$$

B) Controllable Subspace

Given two times $t_0 \leq t_1$ the controllable subspace on $[t_0, t_1]$

$$C[t_0\, t_1] = \{ x_0 \in \mathbb{R}^n \mid \exists\, u(\cdot) \in \mathcal{U}_{[t_0, t_1]},\ 0 = \Phi(t_1, t_0) x_0 + \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) u(\tau)\, d\tau$$

III Reachability Grammian

$$W_R(t_0, t_1) \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) \underbrace{B^T(\tau) \Phi^T(t_1, \tau)}_{u}\, d\tau$$

In this form it looks like our solution to LTV

A) Theorem

$$R[t_0, t_1] = \mathrm{Img}(W_R(t_0, t_1))$$

Moreover: $u(t) = B^T(t) \Phi^T(t_1, \tau) \eta$ where η satisfies

$x_1 \in \mathrm{Im}(M_R) \Rightarrow \exists\, \eta$ s.t

$x_1 = W_R \eta$

$u(t)$ can be used to transfer the state from 0 to x_1

B) Proof

i) $x_1 \in Im(W_R(t_0,t_1)) \Rightarrow x_1 \in R[t_0,t_1]$

Look @ state evolution w/ $u(t) = B^T(t)\Phi^T(t_1,t)\eta_1$

$$x(t_1) = \int_{t_0}^{t_0} \Phi(t_1,\tau)B(\tau)B^T(\tau)\Phi^T(t_1,\tau)\eta_1 \, d\tau$$

$$= W_R(t_0,t_1)\eta_1 = x_1$$

Thus x_1 is in $R[t_0 t_1]$

ii) $x_1 \in R[t_0 t_1] \Rightarrow x_1 \in Im(W_R(t_0,t_1))$

If x_1 is in the reachability set, there must be a control that moves the system to x_1 ← It is in the image space

$x_1 \in R[t_0 t_1] \Rightarrow \exists\, u(\cdot) \in \mathcal{U}[t_0 t_1]$ s.t

$$x_1 = \int_{t_0}^{t_1} \Phi(t_1,\tau)B(\tau)u(\tau)\,d\tau$$

$$\eta_1^T W_R(t_0 t_1)\eta_1 = \int_{t_0}^{t_1} v^T(\tau)v(\tau)\,d\tau$$

$$\Rightarrow = \int_{t_0}^{t_1} \|v(\tau)\|^2$$

$$\Rightarrow \underline{B^T(\tau)\Phi^T(t_1,\tau)\eta_1 = 0}$$

Aside:

If $x^T\eta = 0 \Rightarrow x_1 \in (N(W_R(t_0 t_1))$

$$x_1^T\eta_1 = \int_{t_0}^{t_1} u^T(t)B^T(\tau)\Phi^T(t_1,\tau)\eta_1\,d\tau$$

$$= \int_{t_0}^{t_1} u^T(\tau)\cdot 0 \cdot d\tau = 0$$

$\Rightarrow x_1^T\eta_1 = 0 \Rightarrow x_1 \in (N(W_R(t_0,t_1))^{\perp}$

$\Rightarrow x_1 \in Im(W_R(t_0,t_1))$

IV Controllability Grammian

$$W_c = \int_{t_0}^{t_1} \Phi(t_0, \tau) B(\tau) B^T(\tau) \Phi^T(t_0, \tau) \, d\tau$$

Theorem

$\mathcal{C}(t_0, t_1) = \mathrm{Im}(W_c(t_0, t_1))$

↳ Controllability Subspace

Moreover, the control

$$u(t) = -B^T(t) \Phi^T(t_0, t) \eta_0$$

can be used to move the state from $x(t_0) = x_0$ to $x(t_1) = 0$.

A) Proof (HWK Problem)

i)
- $\exists \eta_0$ s.t. $x_0 = W_c(t_0, t) \eta_0$
- Look at state evolution (use control in the theorem)
- Trick: Left multiply result by $\Phi(t_0, t_1)$
- You will see $x_1 = 0 \Rightarrow x_0 \in \mathcal{C}(t_0, t_1)$

ii)
- Recall $x_0 \in \mathcal{C}(t_0, t_1)$ (write out definition)
- Look at vector $\eta_0 \in N(W_c(t_0, t_1)) \Rightarrow W_c \eta_0 = 0$
- Similarly to before $\eta_0^T W_c \eta_0$
- Use fact that if $x_0^T \eta_0 = 0 \Rightarrow x_0 \in (N(W_c))^\perp$
- Show $x_0^T \eta_0 = 0$

  $\Rightarrow x_0 \in \mathrm{Im}(W_c)$

V Reachability between two Points

We have defind these Grammians from the origin, but what about any 2 points?

Let $d = x_1 - \Phi(t, t_0)$

$\Rightarrow d = \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) u(\tau) d\tau$

Iff $x_1 - \Phi(t, t_0) x_0 \in R(t_0, t)$

then we can move from x_0 to x_1

↳ The same thing as saying

$x_1 - \Phi(t, t_0) x_0 \in Im(WR(t_0, t_1))$

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